Title of the Activity: 8-bit Barrel Shifter

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Description:

A **Barrel Shifter** is a high-speed **combinational digital circuit** used to shift or rotate data bits by a specified number of positions in a single operation. Unlike conventional shift registers that shift data one bit per clock cycle, a barrel shifter performs multi-bit shifting instantly using parallel hardware, making it highly efficient and faster.

The barrel shifter operates based on control inputs that determine the **direction** and **magnitude of shifting**. The direction input specifies whether the shift operation is to the left or right, while the shift amount input defines the number of bit positions by which the data is to be shifted.

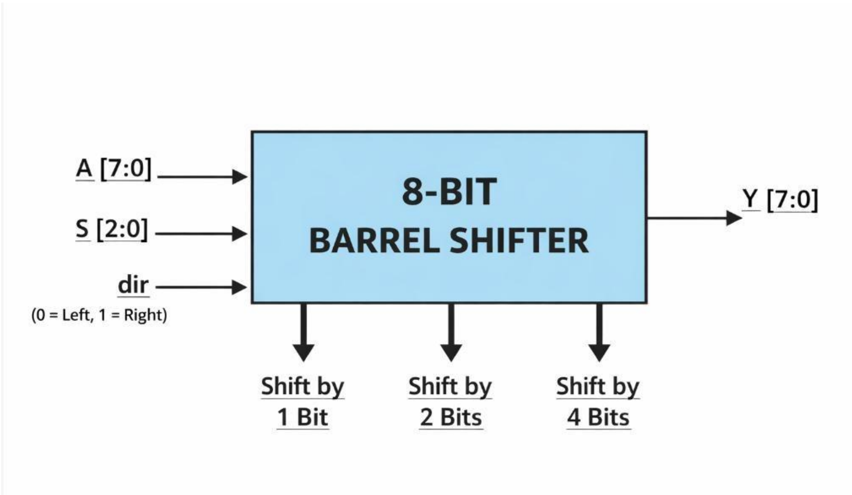
In an 8-bit barrel shifter, the input data consists of 8 bits, and the shift amount is typically represented using 3 bits, allowing shifts from 0 to 7 positions. The circuit produces an output where the bits are shifted accordingly, and vacant positions are filled with zeros in logical shifting.

Internally, the barrel shifter is implemented using a network of **multiplexers arranged in stages**. For an 8-bit design, three stages are commonly used to perform shifts of 1, 2, and 4 bits. By combining these stages, any shift value between 0 and 7 can be achieved efficiently in a single cycle.

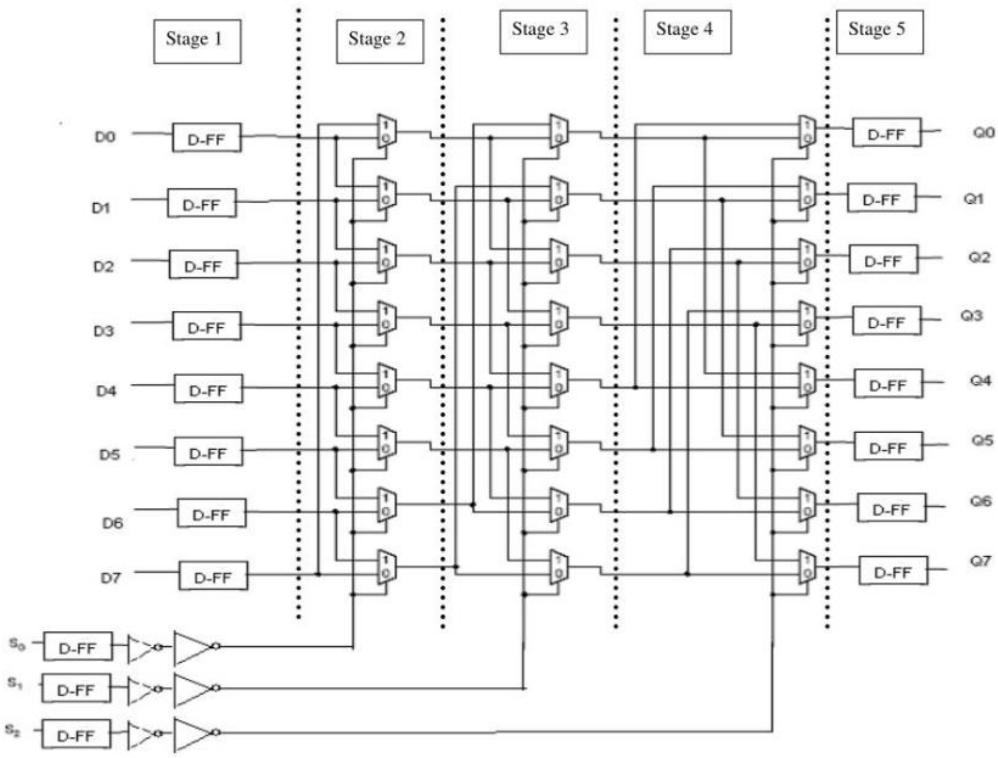
Barrel shifters are widely used in digital systems such as **Arithmetic Logic Units (ALUs), processors, digital signal processing (DSP) units, and data manipulation circuits**, where fast shifting operations are required.

Due to its ability to perform high-speed operations with minimal delay, the barrel shifter plays a crucial role in improving the overall performance of modern digital systems.

Block Diagram:



Logic Diagram



Verilog Code

```
module barrel_shifter(

input [7:0] A,

input [2:0] S,
```

```
input dir,
output reg [7:0] Y
);
always @(*)
begin
case(dir)
1'b0: Y = A << S;  // Left shift
1'b1: Y = A >> S;  // Right shift
endcase
end
endmodule
```

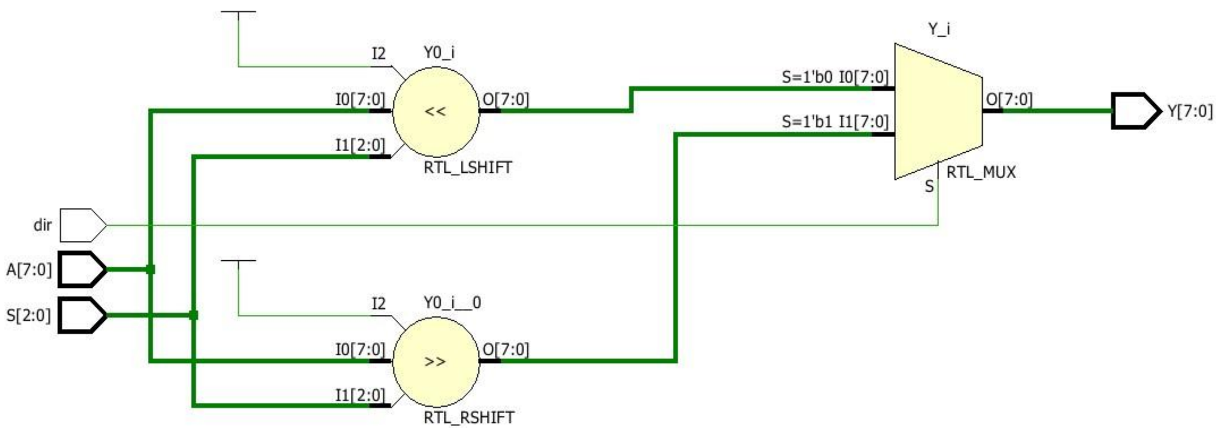
Testbench Code

```
`timescale 1ns/1ps
module barrel_shifter_tb;
reg [7:0] A;
reg [2:0] S;
reg dir;
wire [7:0] Y;
barrel_shifter uut (
   .A(A),
   .S(S),
   .dir(dir),
   .Y(Y)
);
Initial
begin
```

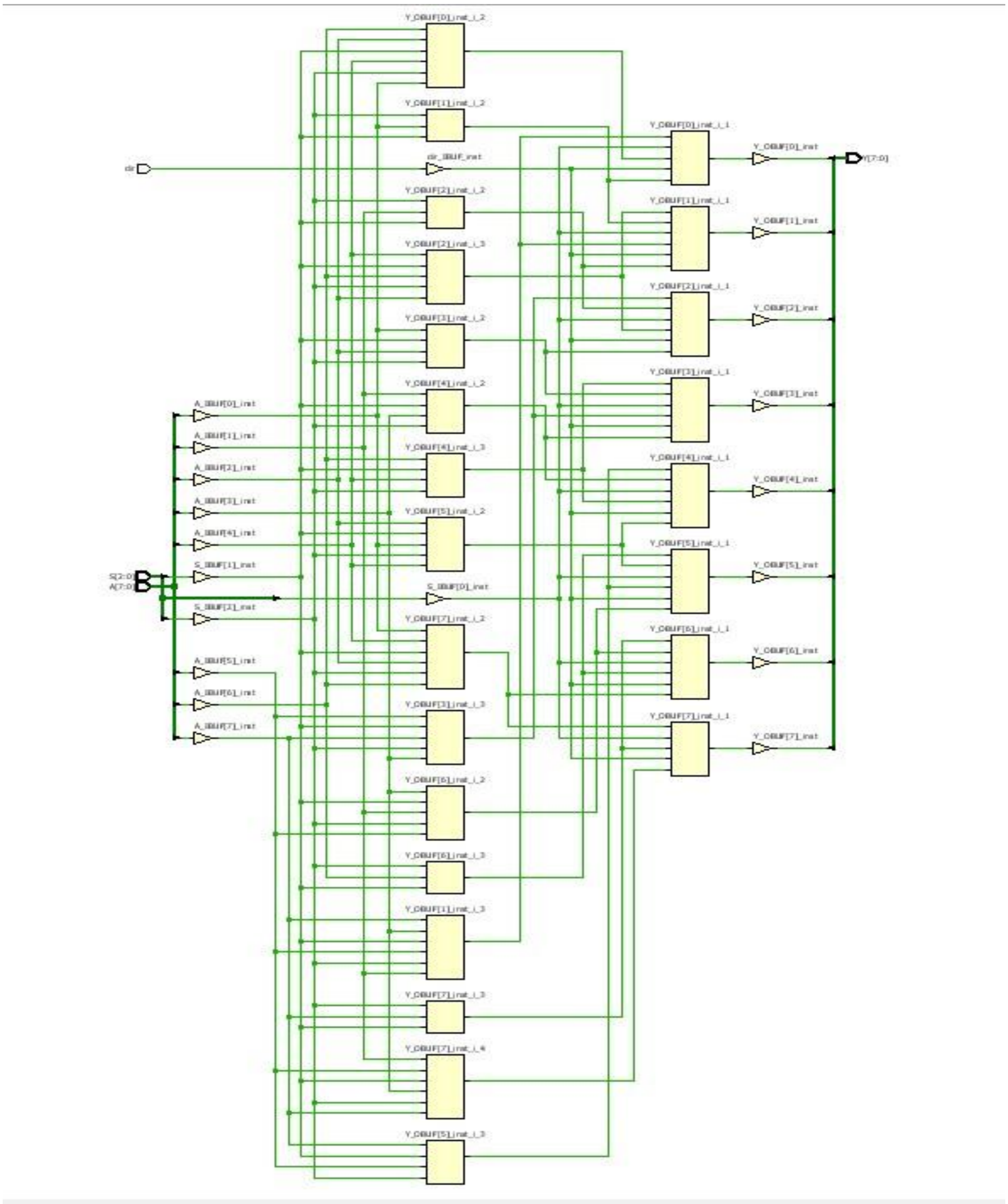
```
A = 8'b10110011;

S = 3'b001;

dir = 0; #10;

S = 3'b010; #10;

dir = 1;

S = 3'b001; #10;

S = 3'b011; #10;

$finish;

end

endmodule
```

Simulation Results

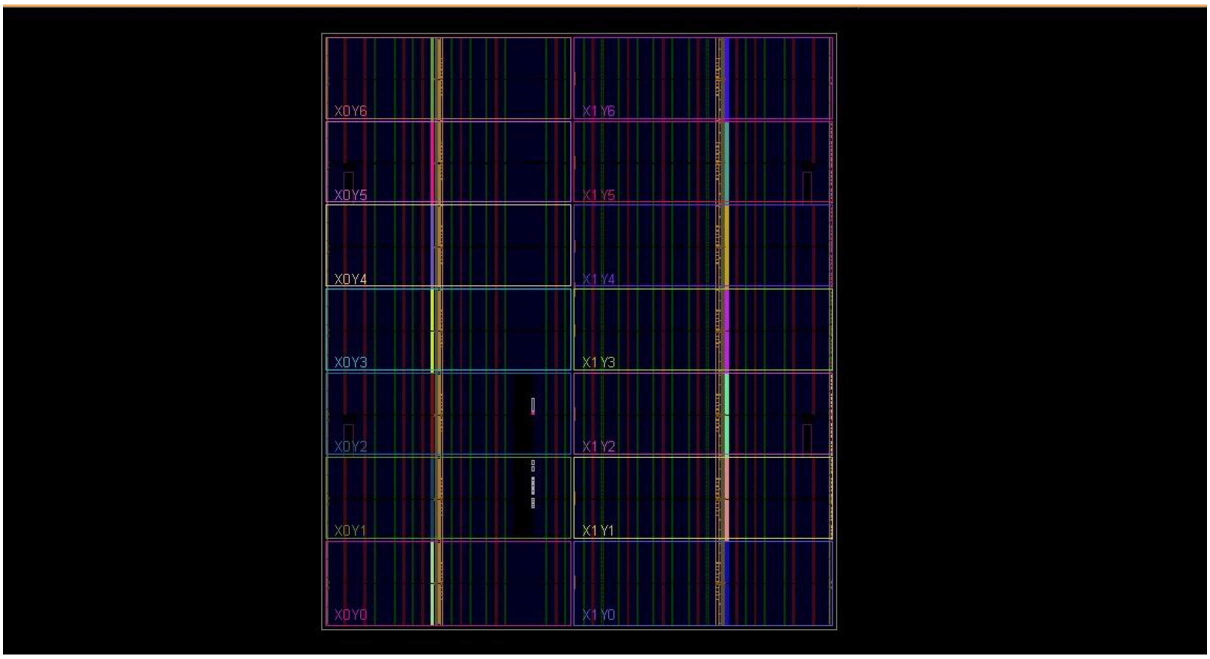
Elaborated Design



Schematic Diagram



Synthesis (RTL Diagram)



Advantages

- High-speed operation
- Performs a shift in one cycle
- Efficient hardware implementation
- Widely used in processors

Applications

- Arithmetic Logic Units (ALU)
- Processors
- DSP systems
- Data manipulation units